METRIC
A-A-59626
27 June 2001
SUPERSEDING
FF-B-171/22
30 November 1993

# COMMERCIAL ITEM DESCRIPTION

## BEARING, BALL, ANNULAR, SINGLE ROW, ANGULAR CONTACT, CONTACT ANGLE 11° THROUGH 22°, DIMENSION SERIES 03

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers metric, single row, angular contact, annular ball bearings for general-purpose use.

2. CLASSIFICATION. The ball bearings shall be classified by the class, sizes, cage materials, shield and seal configurations, precision tolerances, lubricant, preservative, and grease fill requirements, and duplex mounting configurations listed below:

Class 2 - dimension series 03

Size - bearing dimensions (see table I)

Cage materials (see table II)

Shield and seal configurations (see table III)

Precision tolerances (see table IV)

Lubricant, preservative, and grease fill requirements (see table V)

Duplex mounting configuration (see table VI)

3. SALIENT CHARACTERISTICS

3.1 Dimensions. Bearing boundary dimensions (and dynamic load ratings, see 3.4) shall conform to the requirements specified in table I for each of the coded bearing sizes. The listed dimensions conform to the requirements specified for the listed bearing sizes from dimension series 03 in the American National Standards Institute/American Bearing Manufacturers Association (ANSI/ABMA) Standard 20, "Radial Bearings of Ball, Cylindrical Roller, and

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

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Spherical Roller Types, Metric Design" (DoD adopted). The bearing size shall be as specified in the acquisition order (see 7.4(b)). For unlisted bearing sizes, the dimensional and dynamic load rating requirements should also be specified in the acquisition order.

TABLE I. Dimensional requirements (ABMA series 03).

| Bearing size code | Bore diameter (mm) | Outside diameter (mm) | Width (mm) | Chamfer r/min (mm) | Minimum shaft shoulder diameter* (mm) | Dynamic load rating (minimum) (lb) |
|---|---|---|---|---|---|---|
| 01 | 12 | 37 | 12 | 1.0 | 18 | 2,380 |
| 02 | 15 | 42 | 13 | 1.0 | 21 | 2,700 |
| 03 | 17 | 47 | 14 | 1.0 | 23 | 3,330 |
| 04 | 20 | 52 | 15 | 1.1 | 27 | 4,159 |
| 05 | 25 | 62 | 17 | 1.1 | 32 | 5,260 |
| 06 | 30 | 72 | 19 | 1.1 | 37 | 7,010 |
| 07 | 35 | 80 | 21 | 1.5 | 43.5 | 8,520 |
| 08 | 40 | 90 | 23 | 1.5 | 48.5 | 10,409 |
| 09 | 45 | 100 | 25 | 1.5 | 53.5 | 13,200 |
| 10 | 50 | 110 | 27 | 2.0 | 60 | 16,883 |
| 11 | 55 | 120 | 29 | 2.0 | 65 | 19,500 |
| 12 | 60 | 130 | 31 | 2.1 | 72 | 22,300 |
| 13 | 65 | 140 | 33 | 2.1 | 77 | 25,179 |
| 14 | 70 | 150 | 35 | 2.1 | 82 | 27,200 |
| 15 | 75 | 160 | 37 | 2.1 | 87 | 30,799 |
| 16 | 80 | 170 | 39 | 2.1 | 92 | 33,497 |
| 17 | 85 | 180 | 41 | 3.0 | 99 | 36,000 |
| 18 | 90 | 190 | 43 | 3.0 | 104 | 38,892 |
| 19 | 95 | 200 | 45 | 3.0 | 109 | 41,590 |
| 20 | 100 | 215 | 47 | 3.0 | 114 | 44,287 |
| 21 | 105 | 225 | 49 | 3.0 | 119 | 50,132 |
| 22 | 110 | 240 | 50 | 3.0 | 124 | 55,977 |
| 24 | 120 | 260 | 55 | 3.0 | 134 | 59,574 |
| 26 | 130 | 280 | 58 | 4.0 | 148 | 66,094 |
| 28 | 140 | 300 | 62 | 4.0 | 158 | 78,900 |
| 30 | 150 | 320 | 65 | 4.0 | 168 | 78,900 |

* Listed for reference purposes only. Shoulder height shall be determined to provide sufficient clearance for the direct application of bearing removal force against the bearing inner ring. If the required minimum clearance is not available, an alternative non-destructive bearing removal capability shall be provided.



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3.2 Materials.

3.2.1 Rings. The bearing ring material shall be chromium-alloy steel 52100 (UNS G52986) as specified in the American Society for Testing and Materials (ASTM) A 295, "Standard Specification for High-Carbon and Roller Bearing Steel" (DoD adopted). The finished rings shall not exceed the associated billet material inclusion rating also specified in ASTM A 295. Ring hardness shall be at least 58 HRC as defined in ASTM E 18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials" (DoD adopted). The grain size of the ring material shall be determined in accordance with ASTM E 112, "Standard Methods for Determining Average Grain Size" (DoD adopted).

3.2.2 Balls. The bearing ball material shall be chromium-alloy steel 52100 (UNS G52986) as specified in ASTM A 295. The finished balls shall not exceed the associated billet material inclusion rating, which is also specified in ASTM A 295. Balls shall be through-hardened no less than 60 HRC and no more than 67 HRC as defined in ASTM E 18. The grain size of the ball material shall be determined in accordance with ASTM E 112.

3.2.3 Cage. The bearing cage material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. Metallic and non-metallic cages shall meet the same bearing performance requirements. Cage materials shall operate from -65 to 230 °F (-54 to 110 °C). Unless otherwise specified in the acquisition order, the cage material shall be one of the optional coded types listed in table II (see 7.4(c)).

TABLE II. Cage materials.

| Code[1] | Material type |
|---|---|
| J | Pressed steel |
| M | Machined bronze or brass |
| P | Molded plastic (nylon 66 or equal) |
| T | Machined non-metallic (phenolic) |
| Y | Pressed brass |
| X | Other (specify in acquisition order) |

[1] No significant change from FF-B-171/22.

3.2.4 Shields and seals. Unless otherwise specified in the acquisition order, the shield and seal configuration for the bearing shall be one of the coded options listed in table III (see 7.4(d)). The acquisition order shall state which side of the outer ring (heavy shoulder or counterbore side) the shield or seal is to be located (see 7.4(e)).

3.2.4.1 If a single shield or seal is required, the shield or seal material shall be as recommended by the manufacturer, unless otherwise specified in the acquisition order (see 7.4 (f)). The material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. The shield



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or seal material shall remain functionally effective at temperatures ranging from -65 to 230 °F (-54 to 110 °C).

TABLE III. Shield and seal configurations.

| Code[1] | Configuration option |
|---|---|
| A | Open (no shields or seals) |
| X | Other (specify in acquisition order) |

[1] No significant change from FF-B-171/22.

3.3 Precision tolerance. The bearing precision tolerance level shall conform to Annular Bearing Engineers Committee (ABEC) class ABEC-1, ABEC-3, ABEC-5, or ABEC-7 as defined in ANSI/ABMA Standard 20. The tolerance class and associated radial internal clearance shall be one of the coded options listed in table IV (see 7.4(g)).

3.3.1 Bearing precision tolerance levels and internal clearances may be affected by the addition of shields and/or seals. Manufacturers/suppliers should be consulted for availability of desired precision tolerances when shields and/or seals are required.

TABLE IV. Precision tolerance requirements.

| Code | Tolerance class | Radial internal clearance |
|---|---|---|
| A | ABEC-1 | Symbol 2 (less than normal) |
| B | | Symbol 0 (normal) |
| C | | Symbol 3 (greater than normal) |
| D | | Symbol 4 (greater than symbol 3) |
| E | ABEC-3 | Symbol 2 (less than normal) |
| F | | Symbol 0 (normal) |
| G | | Symbol 3 (greater than normal) |
| H | | Symbol 4 (greater than symbol 3) |
| J | ABEC-5 | Symbol 2 (less than normal) |
| K | | Symbol 0 (normal) |
| L | | Symbol 3 (greater than normal) |
| M | | Symbol 4 (greater than symbol 3) |
| N | ABEC-7 | Symbol 2 (less than normal) |
| P | | Symbol 0 (normal) |
| R | | Symbol 3 (greater than normal) |
| S | | Symbol 4 (greater than symbol 3) |

3.4 Dynamic load rating. The listed ratings in table I conform to the requirements specified in ANSI/ABMA Standard 9, "Load Ratings and Fatigue Life for Ball Bearings" (DoD adopted).

3.5 Lubrication, preservation, and grease fill requirements. Unless otherwise specified in the acquisition order, the bearing lubricant or preservative and grease fill requirement shall be as



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selected from the approved coded options listed in table V (see 7.4(h)). The percentage of fill is based on the internal empty space of an assembled bearing.

TABLE V. Lubricant, preservative, and grease fill requirements.

| Code | Lubricant or preservative compound | Percentage (%) of grease fill |
|---|---|---|
| A | Grease in accordance with MIL-PRF-81322 | Up to 25.00 |
| B | | 25.01 - 50.00 |
| C | | 50.01 - 80.00 |
| D | Grease in accordance with DOD-G-24508 | Up to 25.00 |
| E | | 25.01 - 50.00 |
| F | | 50.01 - 80.00 |
| G | Grease in accordance with MIL-PRF-23827 | Up to 25.00 |
| H | | 25.01 - 50.00 |
| J | | 50.01 - 80.00 |
| K | Grease in accordance with SRI-2 or qualified equivalent | Up to 25.00 |
| L | | 25.01 - 50.00 |
| M | | 50.01 - 80.00 |
| S | No fill | 0.00 |
| T | Preservation compound in accordance with MIL-DTL-197 | N/A |
| X | Other (specify in the acquisition order) | As specified |

3.6 Duplex mounting. When required, the adjoining faces of bearings used in duplex applications shall be match ground and marked for proper mounting. The duplex mounting configuration shall be one of the coded options listed in table VI. If required, the duplex mounting configuration shall be specified in the acquisition order (see 7.4(i)).

TABLE VI. Duplex mounting.

| Code | Configuration option |
|---|---|
| F | Face to face |
| B | Back to back |
| T | Tandem |
| U | Universal (F, B, or T) |

3.7 Preload. Preload requirements shall be listed in the acquisition order (see 7.4(j)).

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).



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4.2 Unless otherwise indicated in the solicitation and resulting contract, the foreign acquisition restrictions in Section 225.7019 of the Defense Federal Acquisition Regulation Supplement (DFARS) apply to products described by this CID.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

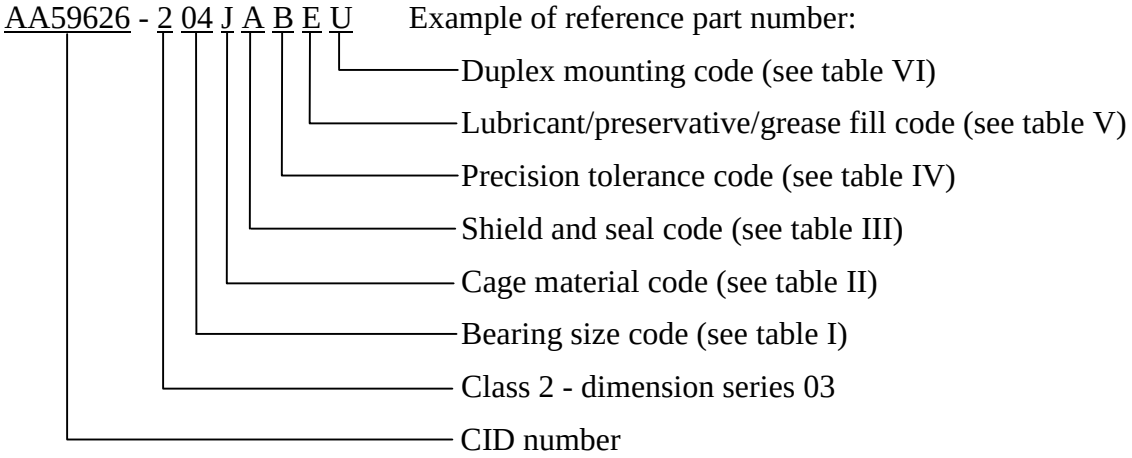
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Unless otherwise specified in the acquisition order, the bearings supplied shall be preserved, packed, and marked in accordance with MIL-DTL-197 (see 7.4(k)).

7. NOTES

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.




AA59626 - 2 04 J A B E U indicates: boundary dimension series 03; bore diameter 20 mm, outside diameter 52 mm, width 15 mm; pressed steel cage; open (no shields or seals); ABEC-1 tolerance class, normal radial internal clearance; filled 25% to 50% with grease in accordance with DOD-G-24508; universal duplex mounting.



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7.2 Sources of documents.

7.2.1 ANSI/ABMA standards. Copies of ANSI/ABMA standards may be obtained from the American Bearing Manufacturers Association, 1200 19th Street NW, Suite 300, Washington, DC 20036-2401.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2.3 Government documents. Copies of FAR, DFARS, and military detail and federal specifications may be obtained from the U.S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328. Electronic copies of military detail specifications and federal specifications may be obtained from http://astimage.daps.dla.mil/quicksearch/.

7.3 Sources of supply. The manufacturers and/or suppliers listed below are known to supply products that meet the salient characteristics requirements of this document. Competition is not limited to the listed firms.

| | |
|---|---|
| FAG Bearings Corporation<br>Danbury, CT 06813 | KOYO Corporation of U.S.A.<br>Westlake, OH 44145 |
| MRC Bearings<br>Jamestown, NY 14702 | NSK Corporation<br>Ann Arbor, MI 48106 |
| NTN Bearing Corporation of America<br>Mt. Prospect, IL 60056 | The Torrington Company<br>Torrington, CT 06790 |

7.4 Ordering data. Acquisition documents shall specify the following information:

a. CID document number, revision, and CID PIN
b. Bearing size (with dimension/load requirements if size is unlisted) (see 3.1)
c. Cage material type (see 3.2.3)
d. Shield and seal configuration (see 3.2.4)
e. Location of shield or seal (see 3.2.4)
f. Shield and seal material (see 3.2.4.1)
g. Precision tolerance requirements (see 3.3)
h. Lubricant/preservative/grease fill requirement (see 3.5)
i. Duplex mounting configuration, if required (see 3.6)
j. Preload requirement, if required (see 3.7)
k. Preservation, packing and marking requirements (see 6.1)

7.5 Codes cross-reference. Tables VII, VIII, and IX contain cross-reference data for the part identification number information as listed in FF-B-171/22 and this CID.

7.5.1 Bearing class designations. The CIDs replacing 33 of the 37 slant sheets of FF-B-171 have been assigned class codes corresponding to ABMA dimension series. Table VII lists the



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FF-B-171 slant sheets, the corresponding dimension series, the CID class codes, and the replacement CIDs.

TABLE VII. Federal specification to CID cross-reference.

| FF-B-171 slant sheets | ABMA dimension series | CID class codes | Replacement CID numbers |
|---|---|---|---|
| 1 | 02, 03, 10 | 1, 2, 4 | A-A-59581 |
| 2 | 19 | 5 | A-A-59582 |
| 3 | 10 | 4 | A-A-59583 |
| 4 | 02 | 1 | A-A-59584 |
| 5 | 03 | 2 | A-A-59585 |
| 6 | 04 | 3 | A-A-59586 |
| 7 | 02 | 1 | A-A-59587 |
| 8 | 03 | 2 | A-A-59589 |
| 9 | 32 | 8 | A-A-59595 |
| 10 | 33 | 9 | A-A-59596 |
| 11 | 02 | 1 | A-A-59597 |
| 12 | 03 | 2 | A-A-59598 |
| 13 | 22 | 6 | A-A-59599 |
| 14 | 23 | 7 | A-A-59600 |
| 15 | 19 | --- | Canceled |
| 16 | 10 | --- | Canceled |
| 17 | 02 | --- | Canceled |
| 18 | 03 | --- | Canceled |
| 19 | 19 | 5 | A-A-59623 |
| 20 | 10 | 4 | A-A-59624 |
| 21 | 02 | 1 | A-A-59625 |
| 22 | 03 | 2 | A-A-59626 |
| 23 | 04 | 3 | A-A-59627 |
| 24 | 10 | 4 | A-A-59628 |
| 25 | 02 | 1 | A-A-59629 |
| 26 | 03 | 2 | A-A-59630 |
| 27 | 04 | 3 | A-A-59631 |
| 28 | 10 | 4 | A-A-59632 |
| 29 | 02 | 1 | A-A-59633 |
| 30 | 03 | 2 | A-A-59634 |
| 31 | 04 | 3 | A-A-59635 |
| 32 | 32 | 8 | A-A-59636 |
| 33 | 33 | 9 | A-A-59637 |
| 34 | 32 | 8 | A-A-59638 |
| 35 | 33 | 9 | A-A-59639 |
| 36 | 32 | 8 | A-A-59640 |
| 37 | 33 | 9 | A-A-59641 |



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TABLE VIII. Radial internal clearance and ABEC tolerance codes.

| FF-B-171/22 codes | | | A-A-59626 codes | | |
|---|---|---|---|---|---|
| Code | Radial internal clearance | Tolerance class | Code | Tolerance class | Radial internal clearance |
| 1 | Symbol 2 | ABEC-1 | A | ABEC-1 | Symbol 2 |
| 2 | Symbol 0 | | B | | Symbol 0 |
| 3 | Symbol 3 | | C | | Symbol 3 |
| 4 | Symbol 4 | | D | | Symbol 4 |
| | | | E | ABEC-3 | Symbol 2 |
| | | | F | | Symbol 0 |
| | | | G | | Symbol 3 |
| | | | H | | Symbol 4 |
| 5 | Symbol 2 | ABEC-5 | J | ABEC-5 | Symbol 2 |
| 6 | Symbol 0 | | K | | Symbol 0 |
| 7 | Symbol 3 | | L | | Symbol 3 |
| 8 | Symbol 4 | | M | | Symbol 4 |
| | | | N | ABEC-7 | Symbol 2 |
| | | | P | | Symbol 0 |
| | | | R | | Symbol 3 |
| | | | S | | Symbol 4 |

TABLE IX. Lubricant, preservative, and grease fill requirements codes.

| FF-B-171/22 codes | | A-A-59626 codes | | |
|---|---|---|---|---|
| Code | Lubricant or preservative | Code | Lubricant or preservative | Grease fill % |
| A | Grease IAW MIL-PRF-81322 | A | Grease IAW MIL-PRF-81322 | Up to 25.00 |
| B | Grease IAW DOD-G-24508 | B | | 25.01 - 50.00 |
| C | Grease IAW MIL-PRF-23827 | C | | 50.01 - 80.00 |
| D | Grease IAW SRI-2 or equivalent | D | Grease IAW DOD-G-24508 | Up to 25.00 |
| E | Preservation compound IAW MIL-DTL-197 | E | | 25.01 - 50.00 |
| | | F | | 50.01 - 80.00 |
| | | G | Grease IAW MIL-PRF-23827 | Up to 25.00 |
| | | H | | 25.01 - 50.00 |
| | | J | | 50.01 - 80.00 |
| | | K | Grease IAW SRI-2 or qualified equivalent | Up to 25.00 |
| | | L | | 25.01 - 50.00 |
| | | M | | 50.01 - 80.00 |
| | | S | No fill | 0.00 |
| | | T | Preservation compound IAW MIL-DTL-197 | N/A |
| | | X | Other (specify in the acquisition | As specified |



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7.6 Subject term (key word) listing.

ABEC
cage
dynamic load rating
lubricant
precision tolerance
rings

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - OS
Air Force - 99

Reviewers:
Army - AR, EA, GL
Navy - SH
Air Force - 11, 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FXE

Preparing activity:
DLA - GS4

(Project 3110-1206)



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